



Grade 7
General
Mathematics



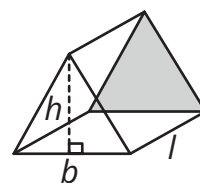
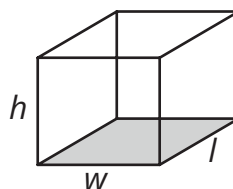
NSCAS NEBRASKA STUDENT-CENTERED ASSESSMENT SYSTEM

Mathematics Reference Sheet

2-Dimensional Shape	Area	Circumference
Circle	$A = \pi r^2$	$C = \pi d = 2\pi r$
Triangle	$A = \frac{1}{2}bh$	Perimeter
Rectangle	$A = l \times w$	$P = 2l + 2w = 2(l + w)$
Square	$A = s \times s$	$P = s + s + s + s$
Trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$	
Parallelogram	$A = bh$	

3-Dimensional Shape	Volume	Surface Area
Rectangular Prism	$V = lwh = Bh$	SA = the sum of the areas of all the faces
Triangular Prism	$V = \frac{1}{2}bhl = Bl$	SA = the sum of the areas of all the faces
Cone	$V = \frac{1}{3}\pi r^2h$	
Cylinder	$V = \pi r^2h$	
Sphere	$V = \frac{4}{3}\pi r^3$	

In the images to the right, the shaded faces are the bases.



2-Dimensional Shape Key		
b = base length	l = length	s = side
h = height	w = width	
d = diameter	r = radius	

Use 3.14 for π .

3-Dimensional Shape Key		
d = diameter	l = length	w = width
r = radius	h = height of shape	B = area of base
	b = base length	

Percent Change	Pythagorean Theorem
% change = $\frac{\text{difference in amounts}}{\text{original amount}}$	$c^2 = a^2 + b^2$

Standard Units	Metric Units
Conversions – Length	
1 foot (ft) = 12 inches (in.)	1 centimeter (cm) = 10 millimeters (mm)
1 yard (yd) = 3 feet (ft) = 36 inches (in.)	1 meter (m) = 100 centimeters (cm)
1 mile (mi) = 1,760 yards (yd) = 5,280 feet (ft)	1 meter (m) = 1,000 millimeters (mm)
	1 kilometer (km) = 1,000 meters (m)
Conversions – Volume	
1 cup = 8 fluid ounces (fl oz)	1 liter (l) = 1,000 milliliters (ml)
1 pint (pt) = 2 cups	1 liter (l) = 1,000 cubic centimeters (cu cm)
1 quart (qt) = 2 pints (pt)	
1 gallon (gal) = 4 quarts (qt)	
Conversions – Weight/Mass	
1 pound (lb) = 16 ounces (oz)	1 gram (g) = 1,000 milligrams (mg)
1 ton = 2,000 pounds (lb)	1 kilogram (kg) = 1,000 grams (g)

Directions:

On the following pages of your practice test booklet are questions for the Grade 7 *Nebraska Student-Centered Assessment System (NSCAS) General Mathematics* practice test.

Read these directions carefully before beginning the test.

This practice test will include several different types of questions. Multiple choice questions will ask you to select an answer from among four choices. Multiple select questions will ask you to select multiple correct answers from among five or more answer choices. For some questions, there may be two parts, Part A and Part B, where each part has a multiple choice or multiple select question. Some questions will ask you to construct an answer by following the directions given.

For all questions:

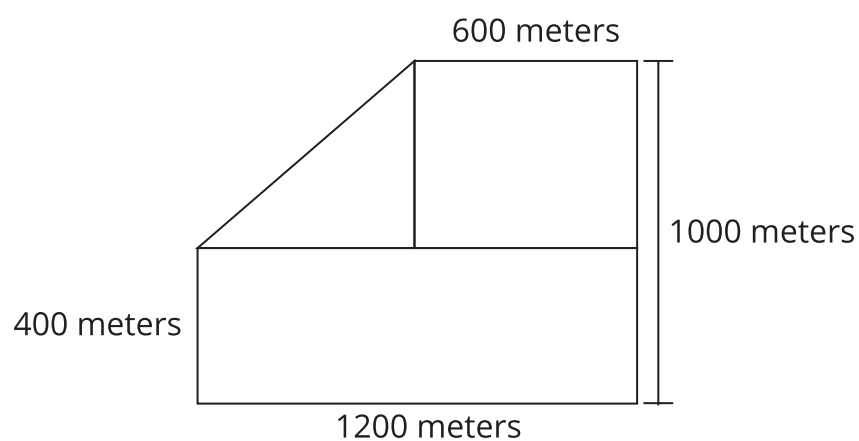
- Read each question carefully and choose the best answer.
- You may use scratch paper to solve the problems.
- The Mathematics Reference Sheet is provided in the back of the test booklet. You may refer to this page at any time during the test.
- Be sure to answer ALL the questions.

1. Which value of x makes this equation true?

$$4x - 6 = -20$$

- A. $x = -\frac{13}{2}$
- B. $x = -\frac{7}{2}$
- C. $x = \frac{7}{2}$
- D. $x = \frac{13}{2}$
2. A cookie jar contains 2 sugar cookies, 4 chocolate chip cookies, and 1 peanut butter cookie. A cookie is chosen, is NOT replaced, and then a second cookie is chosen. Which situations have a probability equal to $\frac{1}{21}$?
Select **all** that apply.
- A. choosing a sugar cookie and then choosing a sugar cookie
- B. choosing a sugar cookie and then choosing a peanut butter cookie
- C. choosing a peanut butter cookie and then choosing a sugar cookie
- D. choosing a peanut butter cookie and then choosing a peanut butter cookie
- E. choosing a chocolate chip cookie and then choosing a peanut butter cookie
- F. choosing a chocolate chip cookie and then choosing a chocolate chip cookie

3. Use the figure below to answer the question.



The figure shows three pieces of land available for development. What is the total area of the land available for development?

- A. 840,000 meters squared
- B. 1,020,000 meters squared
- C. 1,080,000 meters squared
- D. 1,200,000 meters squared

4. Complete the steps used to find the value of $-2\frac{3}{8} \div 5\frac{1}{4}$. Write the numbers in the boxes.

-399 -19 -13 4 8 13 19 21 32 42

$$-2\frac{3}{8} \div 5\frac{1}{4}$$

$$\frac{\boxed{}}{8} \div \frac{\boxed{}}{4}$$

$$\frac{\boxed{}}{\boxed{}} \cdot \frac{\boxed{}}{\boxed{}}$$

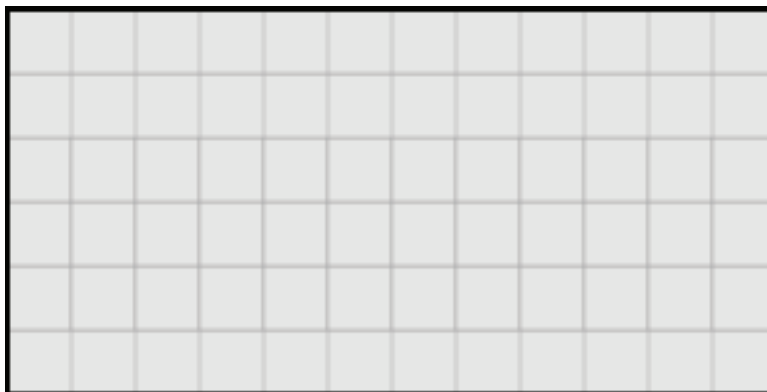
$$\frac{\boxed{}}{\boxed{}}$$

5. What is the value of the expression $2g + h$ when $g = 3.5$ and $h = 7$?

- A. 6.2
- B. 10.7
- C. 14
- D. 21

6. This question has two parts. Answer Part A, and then answer Part B.

Use the diagram to answer the questions.



Joe will build a table using a flat piece of wood for the top. On the scale drawing for the table, a side of 1 grid square represents 6 inches.

Part A

What is the area of the top of the table?

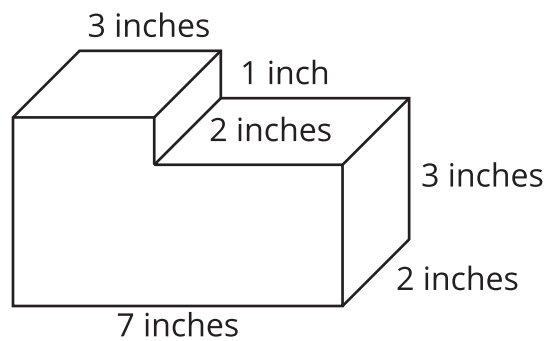
- A. 72 square inches
- B. 216 square inches
- C. 432 square inches
- D. 2,592 square inches

Part B

Joe will use a different type of wood for a border around the top of the table. What is the total length of wood, in inches, Joe needs for the border of the top of the table? Write the answer in the box.

inches

7. Use the diagram of a block below to answer the question.



What is the surface area of the block?

- A. 48 square inches
- B. 92 square inches
- C. 104 square inches
- D. 116 square inches

NSCAS GENERAL — MATHEMATICS

8. The first initials of students' names were placed in a bag. The table shows the letters in the bag.

Letter	Frequency
A	5
C	2
K	3
S	4
W	1

If letters are randomly selected, which events are complementary?

Select **all** that apply.

- A. drawing the letter A
 - B. not drawing the letter P
 - C. drawing the letter C or K
 - D. drawing the letter A, S, or W
 - E. not drawing the letter S or W
9. A laser tag business charges \$99 for a party package plus \$6 per person for equipment. Josiah can spend a maximum of \$175.

Which equation or inequality can Josiah use to determine the number of people who can play laser tag?

- A. $99 + 6x = 175$
- B. $99 + 6x \leq 175$
- C. $99x + 6 = 175$
- D. $99x + 6 \leq 175$

10. What is the value of $-7.52 + 3.01$? Write the answer in the box.

11. As a fundraiser, a student group is selling 2 bags of cookies for \$3.

Which proportion could be used to determine the number of bags of cookies that someone could purchase for \$18?

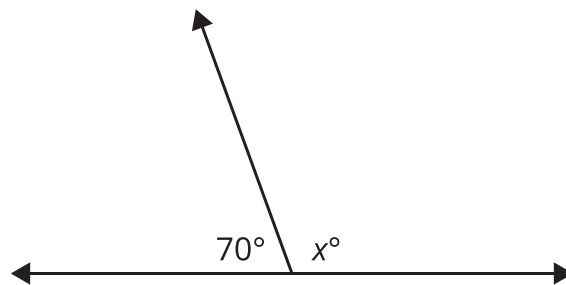
A. $\frac{2}{3} = \frac{x}{18}$

B. $\frac{2}{3} = \frac{18}{x}$

C. $\frac{x}{2} = \frac{3}{18}$

D. $\frac{2}{x} = \frac{18}{3}$

12. Use the figure to complete the sentences.



Select the correct choices to complete the sentences.

The two angles are (**adjacent** or **vertical**) angles.

The value of x is (**70°** **100°** **110°** **180°** **90°**).

13. **This question has two parts. Answer Part A, and then answer Part B.**

Students gathered data about the cars in the teacher parking lot at the school. They counted the number of cars of various colors and recorded the information in a table.

Cars in Teacher Parking Lot

Color	Number of Cars
Black	5
Gray	5
Red	4
Tan	2
White	8
Other	3

Part A

Which question can be answered using the data shown in the table?

- A. How many students like red cars?
- B. Which car color is the most common?
- C. Which car color shows the most dirt?
- D. What is the favorite car color among women?

Part B

The students want to know how many cars of each color in the teacher parking lot were made after a certain year. What additional information do the students need to gather?

- A. the year each car was bought
- B. the year each car was made
- C. the number of years the teachers have taught at the school
- D. the number of years the teachers have been driving the car

14. A store sells all of its products at a price 15% greater than the price the store paid for the product. How much does the store sell a product for when the store paid \$120 for the product?

A. \$102
 B. \$135
 C. \$138
 D. \$180

15. What is the solution to $-3x + 6 \leq 9$? Use the following symbols and numbers to complete the solution.

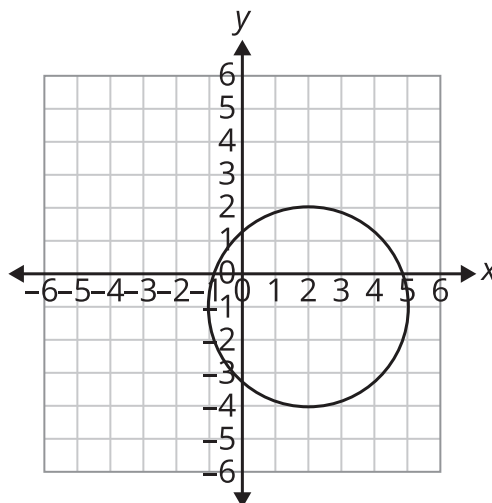
$<$	$\leq$	$>$	$\geq$	-5	-1	1	5
x							

16. What is the value of $-\frac{3}{4}(-2) \div \left(-\frac{2}{3}\right)$?

A. $-\frac{9}{4}$
 B. -1
 C. 1
 D. $\frac{9}{4}$

17. This question has two parts. Answer Part A, and then answer Part B.

Use the graph to answer the questions.



Part A

What is the area of the circle?

- A. 18.84 square units
- B. 19.63 square units
- C. 28.26 square units
- D. 36.00 square units

Part B

What is the circumference of the circle?

- A. 15.70 units
- B. 18.84 units
- C. 28.26 units
- D. 37.68 units

18. Juan spins two different fair spinners. One spinner has numbers 1 through 8. The other has letters A through F. What is the probability that one spinner will land on 3 and the other will land on C?

A. $\frac{1}{48}$

B. $\frac{1}{16}$

C. $\frac{1}{14}$

D. $\frac{1}{7}$

19. Complete the steps to solve for m . Select from the following numbers and write the answers in the boxes.

-28

-21

-4

-3

4

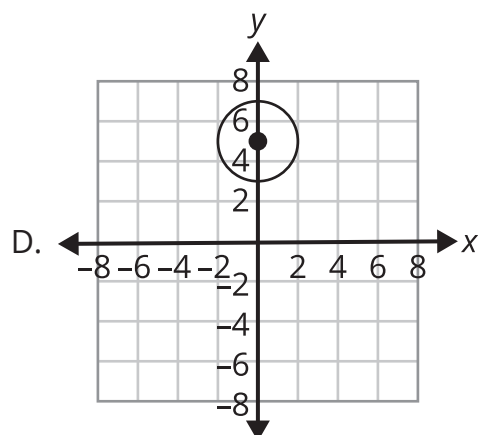
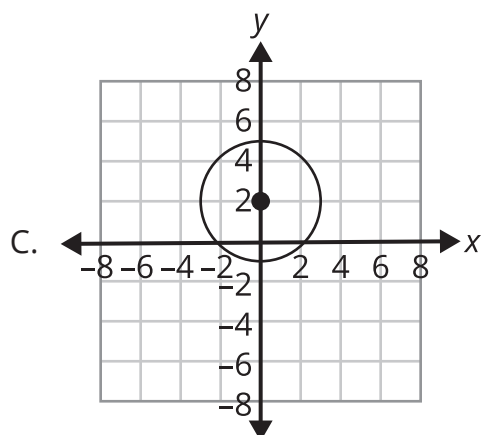
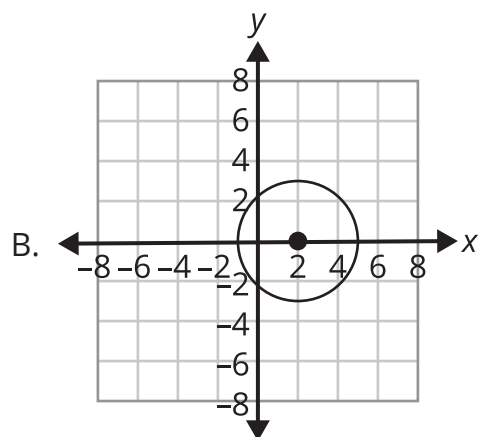
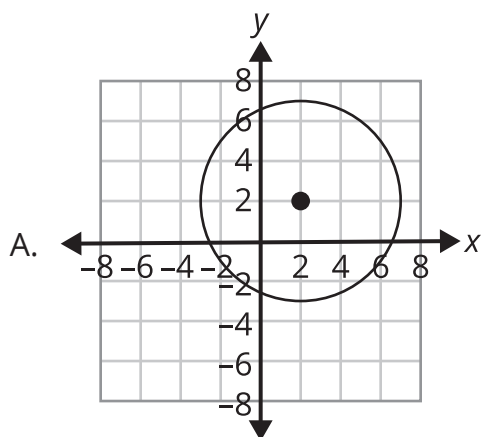
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$$-\frac{3}{4}m + 9 = -12$$

$$-\frac{3}{4}m =$$

$$m =$$

20. Which circle has an area of 25π square units?



**NSCAS General Grade 7
Practice Test Answer Key
Mathematics (English)**



Sequence	Key	Points
1	B	1
2	A, B, C	2
	3 correct, 1 incorrect	1
	2 correct, 1 incorrect	1
	2 correct, 0 incorrect	1
3	B	1
4	$\frac{-19}{8} \div \frac{21}{4}$ $\frac{-19}{8} \cdot \frac{4}{21}$ $\frac{-19}{42}$	1
5	C	1
6	Part A: D Part B: 216 or equivalent	2
	Part A or Part B	1
7	B	1
8	C, D	1
9	B	1
10	-4.51 or equivalent	1
11	A	1
12	adjacent 110°	1
13	Part A: B Part B: B	2
	Part A or Part B	1
14	C	1
15	$x \geq \boxed{-1}$	1
16	A	1
17	Part A: C Part B: B	2
	Part A or Part B	1
18	A	1

NSCAS General Grade 7
Practice Test Answer Key
Mathematics (English)



Sequence	Key	Points
19	$-\frac{3}{4}m = \boxed{-21}$ $m = \boxed{28}$	1
20	A	1